

Test-4

Shrawani Bapu Mule

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1) → ~~What~~ Data is a collection of raw facts, values or information that can be stored and processed. ✓ (1)

2) → ~~What is~~ Database is an organized collection of data that can be stored, accessed, managed and update efficiently. ✓ (1)

3) → Database Management system ✓ (1)

4) → Relational Database Management system ✓ (1)

5) → What is MySQL? ✓ (1)

→ MySQL is an open source-relational database management system (RDBMS) that uses SQL to create, manipulate and manage data. ✓ (1)

6) → What is SQL? ✓ (1)

→ - SQL - structured Query Language
- SQL is a language used to communicate with relational Database. ✓ (1)

7) → Name Any two Database Model? ✓ (1)

→ 1) Hierarchical Database ✓

2) Relational Database ✓

8) What is row in database models?

→ row is a single entry in table

9) What is purpose of the DISTINCT keyword?

→ It will return unique value

10) What does the % wildcard represent in MySQL LIKE?

→ % - percentage symbol represent zero or multiple characters

Section B

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11) Difference between Database and DBMS

Database

DBMS

- | | |
|--|---|
| 1) organized collection of data. | 1) Database Management System |
| 2) can be stored, accessed, managed and update efficiently | 2) can support different database model depending on the system |
| 3) Fast Searching | 3) general database management system. |

12) Differentiate between SQL and MySQL

SQL

MySQL

- | | |
|------------------------|--------------------------------|
| 1) Language | 1) RDBMS / software |
| 2) used to write query | 2) manages Relational Database |
| 3) contain SQL Command | 3) Execute SQL Command |

14) Difference between IN and BETWEEN operators?

IN

BETWEEN

- | | |
|------------------------|----------------------|
| 1) To specify multiple | 1) BETWEEN a certain |
|------------------------|----------------------|

possible values for a column

Range

(2)

2) Select Data with IN

2) To give a range in MySQL we use Between operator

15) Explain IS NULL and NOT NULL with suitable SQL examples.

→ 1) IS NULL

select data is null;
where age is null;

e.g - MySQL select * from employee
→ where address is null;

2) NOT NULL

e.g - select * from employee where age is not null;

13) write any four SQL command categories and give one command for each.

→ 1) DDL - Data Definition language

2) DML - Data Manipulation Language

2) DQL - Data Query Language

4) DCL - Data control Language

(1)

Section C

16) Explain DBMS and RDBMS at least 3

→ 1) DBMS

- Database Management System
- General Database Management system
- Can support different database model depending on the system.

2) RDBMS

- Relational Database Management system
- Based on relational model
- Data represented using table

17) Explain the following database models with their basic structure.

→ 1] Hierarchical Database

- A hierarchical database organizes data in tree-like structure

parent
|

+ --- child

2] Network Database

- A network database organizes data as interconnected records and supports

multiple relationships among records.

student

course Teacher

3] Relational Database

— A relational database organizes data into tables (relations) consisting rows and columns.

student_id name Age course_id

101 shrawani 21 col

102 siddhi 22 CO2

4] object-oriented database

— An object-oriented database stores data in the form of objects.

e.g:—

student object

+ name = shrawani

+ age = 21

+ course = B.Tech

5] NoSQL Database

- NoSQL is broad category of database that uses data models other than, or beyond the traditional-table model.

18] Explain the six MySQL constraints given in the notes. Write the purpose of each!

→ 1) NOT NULL

- ~~It~~ a column cannot accept a NULL or empty value.

2) Unique

- All values in a column are completely distinct from one another

3) DEFAULT

- All values entered into a column satisfy a specific Boolean condition.

4) Check

- All values entered into a column satisfy a Boolean condition.

5) primary key

— uniquely identifies each record in a table.

6) primary key ✓

— A column that links to a primary key in another table.

